THE RANKERS ACADEMY — NAGPUR

BIOLOGY TEACHING PLAN (NEET 2028) | 95 Content Lectures | DPP 1–95 | Each Lecture: 45 Minutes

1 Biology lecture per teaching day (Mon–Fri) | Saturday = DPP discussion + doubt session | NCERT is the BIBLE for NEET

Chapter 1: The Living World | Apr 27 – Apr 30 | 4 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L1	What is living? Characteristics of life	Growth, reproduction, metabolism, consciousness; difference between living & non-living; self-replication	DPP 1	45 min
L2	Biodiversity & nomenclature	Biodiversity definition; need for classification; binomial nomenclature rules; ICBN, ICZN, ICVN	DPP 2	45 min
L3	Taxonomic categories & tools	Species, genus, family, order, class, phylum, kingdom, domain; herbarium, botanical garden, museum, zoological park, key	DPP 3	45 min
L4	The Living World PYQ drill	NEET 2019-2025 PYQs on nomenclature rules; taxonomic hierarchy; identification vs classification	DPP 4	45 min

Chapter 2: Biological Classification | May 1 – May 8 | 6 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L5	Two-Kingdom to Five-Kingdom classification	Whittaker's criteria: cell structure, body organization, nutrition, reproduction, phylogeny; comparison table	DPP 5	45 min
L6	Kingdom Monera – Bacteria	Cell structure; shapes (cocci, bacilli, spirilla, vibrio); Gram +/−; nutrition types; reproduction; archaebacteria vs eubacteria	DPP 6	45 min
L7	Kingdom Monera – Cyanobacteria & Mycoplasma	Cyanobacteria (BGA): heterocyst, nitrogen fixation; Mycoplasma: smallest, no cell wall; chemosynthetic bacteria	DPP 7	45 min
L8	Kingdom Protista	Chrysophytes (diatoms, desmids); dinoflagellates (red tide); euglenoids; slime moulds; protozoans (4 groups)	DPP 8	45 min
L9	Kingdom Fungi	General characters; classification: Phycomycetes, Ascomycetes, Basidiomycetes, Deuteromycetes; lichens; mycorrhiza	DPP 9	45 min
L10	Viruses, Viroids & Prions + PYQ drill	Virus structure (TMV, bacteriophage); DNA vs RNA viruses; viroids (PSTVd); prions; NEET PYQ drill	DPP 10	45 min

Chapter 3: Plant Kingdom | May 11 – May 20 | 8 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L11	Algae – classification & characters	Chlorophyceae (green), Phaeophyceae (brown), Rhodophyceae (red); pigments; storage food; cell wall; reproduction	DPP 11	45 min
L12	Bryophytes	General characters; amphibians of plant kingdom; Liverworts (Marchantia); Mosses (Funaria); alternation of generations	DPP 12	45 min
L13	Pteridophytes	General characters; first vascular plants; homosporous vs heterosporous; Selaginella, Equisetum, ferns; life cycle	DPP 13	45 min
L14	Gymnosperms	General characters; naked seeds; Cycas, Pinus, Ginkgo, Gnetum; male & female cones; life cycle overview	DPP 14	45 min
L15	Angiosperms – overview	Defining features; double fertilization; monocots vs dicots comparison table; economic importance	DPP 15	45 min
L16	Plant life cycles & alternation of generations	Haplontic, diplontic, haplo-diplontic cycles; comparison across plant groups; dominant generation in each	DPP 16	45 min

L17	Plant Kingdom comparison table + diagrams	Complete comparison: Algae vs Bryophyte vs Pteridophyte vs Gymnosperm vs Angiosperm; key NCERT diagrams	DPP 17	45 min
L18	Plant Kingdom PYQ drill	NEET 2019-2025 PYQs; life cycle matching; classification traps; NCERT exact line questions	DPP 18	45 min

Chapter 4: Animal Kingdom | Jun 8 – Jun 19 | 10 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L19	Basis of classification	Levels of organisation (cellular, tissue, organ, organ system); symmetry; diploblastic vs triploblastic; coelom types	DPP 19	45 min
L20	Phylum Porifera & Cnidaria	Porifera: canal system, spicules, spongin; Cnidaria: cnidoblasts, polymorphism, corals; examples with diagrams	DPP 20	45 min
L21	Phylum Ctenophora & Platyhelminthes	Ctenophora: comb plates, bioluminescence; Platyhelminthes: flatworms, flame cells, Taenia, Fasciola, Planaria	DPP 21	45 min
L22	Phylum Aschelminthes & Annelida	Aschelminthes: roundworms, pseudocoelom, Ascaris, Wuchereria; Annelida: segmentation, Nereis, earthworm, leech	DPP 22	45 min
L23	Phylum Arthropoda & Mollusca	Arthropoda: largest phylum, chitinous exoskeleton, classes (Crustacea, Insecta, Arachnida); Mollusca: mantle, radula, Pila, Octopus	DPP 23	45 min
L24	Phylum Echinodermata & Hemichordata	Echinodermata: water vascular system, tube feet, radial symmetry (adults); Hemichordata: Balanoglossus, stomochord	DPP 24	45 min
L25	Phylum Chordata – Protochordata & Cyclostomata	Notochord, dorsal nerve cord, pharyngeal gill slits; Urochordata, Cephalochordata; Cyclostomata: Petromyzon, Myxine	DPP 25	45 min
L26	Pisces, Amphibia & Reptilia	Chondrichthyes vs Osteichthyes; Amphibia: dual life, 3-chambered heart; Reptilia: 3-chambered (croc 4), scales	DPP 26	45 min
L27	Aves & Mammalia	Aves: feathers, pneumatic bones, oil gland, 4-chambered heart; Mammalia: mammary glands, hair, diaphragm, viviparity	DPP 27	45 min
L28	Animal Kingdom PYQ drill	NEET 2019-2025 PYQs; phylum identification; example matching; symmetry & coelom traps; NCERT exact lines	DPP 28	45 min

Chapter 5: Morphology of Flowering Plants | Jun 22 – Jun 30 | 7 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L29	Root system	Tap root vs fibrous root; regions of root; modifications: storage (carrot), pneumatophores, prop, stilt, parasitic	DPP 29	45 min
L30	Stem – types & modifications	Types of stems; modifications: underground (rhizome, tuber, bulb, corm); aerial (tendril, thorn, phylloclade)	DPP 30	45 min
L31	Leaf – morphology & modifications	Parts of leaf; venation; simple vs compound (pinnate, palmate); phyllotaxis; modifications: tendril, spine, pitcher	DPP 31	45 min
L32	Inflorescence & flower	Racemose vs cymose; types; flower parts: calyx, corolla, androecium, gynoecium; actinomorphic vs zygomorphic	DPP 32	45 min
L33	Fruit & seed	True vs false fruit; simple, aggregate, composite fruits; seed structure: monocot vs dicot; endosperm	DPP 33	45 min
L34	Floral formulas & family descriptions	Fabaceae (pea), Solanaceae (potato), Liliaceae (lily); floral formula notation; floral diagram; economic importance	DPP 34	45 min
L35	Morphology PYQ drill	NEET 2019-2025 PYQs; modification identification; family matching; floral formula; NCERT diagram-based Qs	DPP 35	45 min

Chapter 6: Anatomy of Flowering Plants | Jul 1 – Jul 9 | 7 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration

L36	Tissues – meristematic	Apical, lateral, intercalary meristems; characteristics; dedifferentiation & redifferentiation	DPP 36	45 min
L37	Simple permanent tissues	Parenchyma (chlorenchyma, aerenchyma); collenchyma (angular, lamellar, lacunar); sclerenchyma (fibres, sclereids)	DPP 37	45 min
L38	Complex permanent tissues	Xylem: tracheids, vessels, xylem parenchyma, xylem fibres; Phloem: sieve tubes, companion cells, phloem parenchyma	DPP 38	45 min
L39	Anatomy of dicot & monocot root	Epidermis, cortex, endodermis (Casparian strip), pericycle, vascular bundles; dicot vs monocot root comparison	DPP 39	45 min
L40	Anatomy of dicot & monocot stem	Dicot: open vascular bundles, cambium present; Monocot: closed, scattered bundles; detailed comparison	DPP 40	45 min
L41	Anatomy of leaf & secondary growth	Dorsiventral (dicot) vs isobilateral (monocot) leaf; secondary growth: vascular cambium, cork cambium, annual rings	DPP 41	45 min
L42	Anatomy PYQ drill	NEET 2019-2025 PYQs; tissue identification from diagrams; dicot vs monocot; secondary growth; NCERT exact lines	DPP 42	45 min

Chapter 7: Structural Organisation in Animals | Jul 10 – Jul 15 | 4 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L43	Animal tissues – epithelial	Simple epithelium (squamous, cuboidal, columnar, ciliated); compound epithelium; glandular epithelium; cell junctions	DPP 43	45 min
L44	Connective, muscular & neural tissues	Connective: loose, dense, specialised (cartilage, bone, blood); Muscular: skeletal, smooth, cardiac; Neural tissue	DPP 44	45 min
L45	Cockroach morphology & anatomy	Head, thorax, abdomen; digestive system (hepatic caeca, Malpighian tubules); circulatory (haemocoel); respiratory (spiracles, tracheae)	DPP 45	45 min
L46	Structural Org. PYQ drill	NEET 2019-2025 PYQs; tissue identification; cockroach anatomy; matching tissue to function; NCERT diagrams	DPP 46	45 min

Chapter 8: Cell: The Unit of Life | Jul 27 – Aug 4 | 7 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L47	Cell theory & cell types	Cell theory (Schleiden, Schwann, Virchow); prokaryotic vs eukaryotic cells; plant vs animal cell comparison	DPP 47	45 min
L48	Cell envelope & membrane	Cell wall (plant); plasma membrane: fluid mosaic model, transport (passive, active, osmosis); membrane proteins	DPP 48	45 min
L49	Endomembrane system	ER (smooth, rough); Golgi apparatus (cis, trans face); lysosomes (polymorphic); vacuoles	DPP 49	45 min
L50	Mitochondria & plastids	Mitochondria: structure, matrix, cristae, own DNA, semi-autonomous; Plastids: chloroplast, chromoplast, leucoplast	DPP 50	45 min
L51	Ribosomes, cytoskeleton & centrioles	Ribosomes (70S vs 80S); cytoskeleton (microtubules, microfilaments, intermediate); cilia & flagella (9+2); centriole	DPP 51	45 min
L52	Nucleus	Nuclear envelope, nucleoplasm, chromatin, nucleolus; chromosome structure, types; karyotype; euchromatin vs heterochromatin	DPP 52	45 min
L53	Cell PYQ drill	NEET 2019-2025 cell PYQs; organelle function matching; membrane transport; NCERT diagram-based Qs	DPP 53	45 min

Chapter 9: Biomolecules | Aug 5 – Aug 12 | 6 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L54	Carbohydrates	Monosaccharides (glucose, fructose); disaccharides (sucrose, lactose, maltose); polysaccharides (starch, glycogen, cellulose, chitin)	DPP 54	45 min

L55	Proteins	Amino acids (20 standard); peptide bond; primary, secondary (α-helix, β-sheet), tertiary, quaternary structure; denaturation	DPP 55	45 min
L56	Lipids & nucleic acids	Lipids: fats, phospholipids, steroids; Nucleic acids: DNA vs RNA; nucleotide structure; Watson-Crick model; Chargaff's rule	DPP 56	45 min
L57	Enzymes – classification & properties	Enzyme nomenclature; 6 classes (oxidoreductases, transferases, etc.); active site; substrate specificity; lock-and-key vs induced fit	DPP 57	45 min
L58	Enzyme kinetics & factors	Michaelis-Menten (qualitative); Km, Vmax; effect of temperature, pH, substrate conc, inhibitors (competitive, non-competitive)	DPP 58	45 min
L59	Biomolecules PYQ drill	NEET 2019-2025 PYQs; enzyme classification; carbohydrate identification; protein structure; NCERT exact lines	DPP 59	45 min

Chapter 10: Cell Cycle & Cell Division | Aug 13 – Aug 19 | 5 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L60	Cell cycle phases	G1, S, G2 (interphase); M phase; duration; G0 (quiescent); cyclins & CDKs (brief); checkpoints	DPP 60	45 min
L61	Mitosis	Prophase, metaphase, anaphase, telophase – detailed events; cytokinesis (plant vs animal); significance of mitosis	DPP 61	45 min
L62	Meiosis I	Prophase I (leptotene, zygotene, pachytene, diplotene, diakinesis); crossing over; metaphase I, anaphase I, telophase I	DPP 62	45 min
L63	Meiosis II & significance	Meiosis II similar to mitosis; significance: genetic variation, gamete formation, maintains chromosome number	DPP 63	45 min
L64	Cell Division PYQ drill	NEET 2019-2025 PYQs; mitosis vs meiosis comparison; prophase I stages; chromosome number problems; NCERT diagrams	DPP 64	45 min

Chapter 11: Transport in Plants | Aug 20 – Aug 25 | 4 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L65	Means of transport & water potential	Diffusion, facilitated diffusion, active transport, osmosis; water potential ($\Psi = \Psi s + \Psi p$); plasmolysis; imbibition	DPP 65	45 min
L66	Absorption & ascent of sap	Root hair absorption; apoplast & symplast pathways; Casparian strip role; transpiration pull (cohesion-tension theory)	DPP 66	45 min
L67	Transpiration & translocation	Transpiration types (stomatal, lenticular, cuticular); factors; guttation; translocation: pressure flow hypothesis (Münch)	DPP 67	45 min
L68	Transport PYQ drill	NEET 2019-2025 PYQs; water potential problems; ascent of sap theories; translocation; NCERT exact statements	DPP 68	45 min

Chapter 12: Mineral Nutrition | Aug 26 – Aug 28 | 3 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L69	Essential minerals & criteria	Criteria for essentiality (Arnon); macronutrients & micronutrients; role of each element; deficiency symptoms	DPP 69	45 min
L70	Nitrogen metabolism	Nitrogen cycle; biological nitrogen fixation (nitrogenase, Rhizobium, leghaemoglobin); ammonification, nitrification, denitrification	DPP 70	45 min
L71	Mineral Nutrition PYQ drill	NEET 2019-2025 PYQs; deficiency symptom matching; nitrogen fixation; essential element roles; NCERT lines	DPP 71	45 min

Chapter 13: Photosynthesis | Sep 1 – Sep 8 | 6 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration

L72	Light reactions – photosystems	Photosystem I & II; reaction centres (P680, P700); Z-scheme; non-cyclic & cyclic photophosphorylation; chemiosmosis	DPP 72	45 min
L73	Calvin cycle (C3 pathway)	Carbon fixation (RuBisCO); reduction; regeneration of RuBP; 3 turns = 1 G3P; net equation; ATP & NADPH usage	DPP 73	45 min
L74	C4 pathway & CAM	C4 plants: Kranz anatomy, PEP carboxylase, bundle sheath; CAM plants: temporal separation; photorespiration	DPP 74	45 min
L75	Factors affecting photosynthesis	Blackman's law of limiting factors; light intensity, CO concentration, temperature, water; light compensation point	DPP 75	45 min
L76	Chloroplast structure & pigments	Chloroplast ultrastructure; pigments: chlorophyll a, b; carotenoids; xanthophylls; absorption & action spectra; Engelmann's experiment	DPP 76	45 min
L77	Photosynthesis PYQ drill	NEET 2019-2025 PYQs; Z-scheme; Calvin cycle steps; C3 vs C4 comparison; pigment absorption; NCERT diagrams	DPP 77	45 min

Chapter 14: Respiration in Plants | Sep 9 – Sep 14 | 4 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L78	Glycolysis	Steps (10 reactions – overview); substrate level phosphorylation; net gain: 2 ATP, 2 NADH, 2 pyruvate; occurs in cytoplasm	DPP 78	45 min
L79	Krebs cycle & ETC	Krebs cycle: acetyl CoA entry, 8 steps overview, products per turn; ETC: complexes I-IV, chemiosmosis, 34 ATP; total = 36/38 ATP	DPP 79	45 min
L80	Fermentation & respiratory quotient	Anaerobic respiration: alcohol & lactic acid fermentation; RQ definition; RQ values for carbs, fats, proteins, organic acids	DPP 80	45 min
L81	Respiration PYQ drill	NEET 2019-2025 PYQs; ATP yield; glycolysis vs Krebs; RQ calculation; ETC complexes; NCERT exact statements	DPP 81	45 min

Chapter 15: Plant Growth & Development | Sep 21 – Sep 25 | 4 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L82	Growth phases & measurement	Phases: lag, log (exponential), stationary, senescence; growth rate (arithmetic vs geometric); growth curve; conditions for growth	DPP 82	45 min
L83	Plant hormones – Auxin, Gibberellin, Cytokinin	Auxin: discovery, effects (apical dominance, phototropism); GA: bolting, fruit growth; Cytokinin: cell division, delay senescence	DPP 83	45 min
L84	ABA, Ethylene & Photoperiodism	ABA: stress hormone, stomatal closure; Ethylene: fruit ripening, senescence; Photoperiodism: SDP, LDP, DNP; vernalization	DPP 84	45 min
L85	Plant Growth PYQ drill	NEET 2019-2025 PYQs; hormone function matching; photoperiodism; vernalization; growth curve; NCERT statements	DPP 85	45 min

Chapter 16: Digestion & Absorption | Sep 26 – Oct 1 | 4 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L86	Alimentary canal & digestive glands	Mouth to anus: buccal cavity, oesophagus, stomach (cardiac, fundic, pyloric), small intestine, large intestine; liver, pancreas, salivary glands	DPP 86	45 min
L87	Digestion process	Salivary amylase → maltose; pepsin (HCl, pepsinogen); trypsin, chymotrypsin; lipase; bile salts; brush border enzymes; absorption in ileum	DPP 87	45 min
L88	Absorption & disorders	Absorption mechanisms; villi & microvilli; assimilation; egestion; disorders: PEM (kwashiorkor, marasmus), indigestion, constipation, jaundice	DPP 88	45 min
L89	Digestion PYQ drill	NEET 2019-2025 PYQs; enzyme-substrate matching; absorption site; disorders; NCERT diagram-based; caloric value	DPP 89	45 min

Chapter 17: Breathing & Exchange of Gases | Oct 2 – Oct 6 | 3 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L90	Respiratory system & mechanism	Human respiratory system: nose, larynx, trachea, bronchi, lungs, alveoli; mechanism: inspiration & expiration; lung volumes & capacities	DPP 90	45 min
L91	Gas transport & regulation	O_2 transport: oxyhaemoglobin, O_2 dissociation curve, Bohr effect; CO_2 transport: bicarbonate, carbamino, dissolved; regulation of breathing	DPP 91	45 min
L92	Breathing PYQ drill	NEET 2019-2025 PYQs; lung volumes; O_2 dissociation curve; CO_2 transport; disorders (asthma, emphysema); NCERT lines	DPP 92	45 min

Chapter 18: Body Fluids & Circulation + Excretion | Oct 7 – Oct 14 | 6 Lectures

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
L93	Blood composition & coagulation	Plasma, RBC, WBC (types), platelets; blood groups (ABO, Rh); coagulation cascade (simplified); lymph	DPP 93	45 min
L94	Heart & circulatory system	Heart anatomy: 4 chambers, valves, SA node, AV node, Bundle of His, Purkinje; cardiac cycle; ECG waves (P, QRS, T)	DPP 94	45 min
L95	Double circulation & disorders + Excretion intro	Pulmonary vs systemic circulation; cardiac output; disorders (hypertension, CAD, heart failure); Excretory organs overview	DPP 95	45 min

Phase 5–7: Biology Completion & Revision | Oct 22 – Jan 15 | 57 Slots

L#	Topic	Sub-topics / Key Points to Cover	DPP	Duration
—	Oct 22–Nov 4: Excretion + Locomotion complete	Excretory Products (nephron, urine formation, osmoregulation); Locomotion & Movement (skeletal, muscular systems); PYQ drill	DPP 96–105	45 min
—	Nov 12–Nov 20: Neural + Chemical Coordination	Neural Control (neuron, synapse, brain, reflex arc); Chemical Coordination (endocrine glands, hormones, feedback); PYQ drill	DPP 106–115	45 min
—	Nov 21–Dec 19: Full Revision Cycle 1 + NCERT	All 22 chapters rapid revision; NCERT line-by-line reading (Biology = NCERT exam); diagram practice; PYQ drills	Rev DPPs	45 min
—	Dec 21–Jan 15: Mock Analysis + Weak Ch + NCERT	3 full mocks/week; Human Physiology deep drill (highest NEET weightage); diagram drawing practice; NCERT final read	Mock papers	45 min